

What is ML? Why does ML matter? ML: study of algorithms (learn from data). Learn from patterns → make decisions or generate content. ML as adaptive systems - across industries (healthcare, recommendation, ai assistants)

Program VS Learning: Rules vs data-driven learning.

Traditional Programming	Machine Learning	Modern AI
Rules → Program → Output	Data → Models → Predictions	Pretrained Model → Adapt → Deploy

AI system pipeline:

Collect data → preprocess data (why? Noisy, outlier) - not confused models → training → evaluation → deployment → monitoring.

Type of ML:

Supervised learning: predict outputs from labeled data. Regression: predict continuous value: price, temperature (numeric value). Classification: Classify images into cancer/non-cancer. (categories)
Unsupervised learning: (used when labels are unavailable). Find hidden patterns in unlabelled data. (clustering, anomaly detection).
Deep Learning: Neural networks learn complex patterns. CNN (images), RNN → Sequences, Transformer → Language
Reinforcement Learning. Self-supervised learning:
Embeddings: numeric representations of data. (search, recommendation, clustering, and retrieval)
Foundation Models: Large Pretrained Models (GPT, Llama, Claude, Stable Diffusion). Trained once on a massive dataset, and can be adapted for many different tasks (Gemini, ChatGPT, Stable Discussion).
GenAI: The model creates content (text, code, images, audio, video). Application: chatbots, copilots, design tools. ML vs Statistics vs AI. Different goals. Statistics → inference, machine learning → prediction, AI → decision systems. [Modern AI combines all three. Important issues: bias, fairness, privacy, hallucinations, safety. (understanding risks is critical). Trends: AI Agent, multimodal models, scientific discovery, human-ai collaboration → AI literacy will be essential.

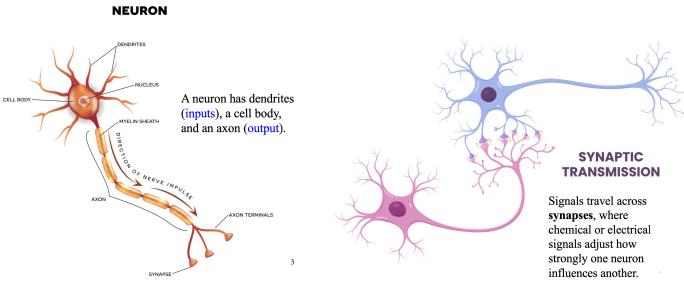
Introduction to ANN Part I

Biological Neuron → Artificial Neuron

Q: Explain how the structure of a biological neuron maps to an artificial neuron. Which aspects of the analogy are most useful for understanding modern neural networks, and where does the analogy break down?

Neuron has dendrites (multiple inputs), a cell body, and axon (output). Signals travel across synapses (where chemical or electrical signals adjust how strongly one neuron influences each other.

- A single neuron does not talk to just one other neuron. But it can send signals to thousands at once.
- Each signal can be weak, strong, turned up, or turned down. Together, this creates a dense web of communication.



From biological neuron → artificial neuron. (mapping the analogy)

1. Dendrites → correspond to inputs (numbers entering the model).
2. Synapses → correspond to Weights (how strongly each input matters).
3. Cell body → correspond to (summation + activation function)
4. Axon → correspond to output of neuron (many simple units connected into large networks)

The useful part? Learning happens by changing connection strength. Many simple units working together can solve complex problems. Information flows through connected layers of neurons. Where? Real neurons are much more complex than artificial neurons. Biological neurons send electrical spikes over time, but artificial intelligence only passes a single number.

Q: Neural networks rely on many weighted connections between units. How does adjusting these weights during training relate to learning in biological systems, and why is this adjustment central to model performance?

- How? In the brain, neurons connect with synapses. Once we learn something, the strength of those synapses connections changes. Artificial neural networks mirror this: weights are synapses. High weight → high connection, near-zero, low connection.
- Why?

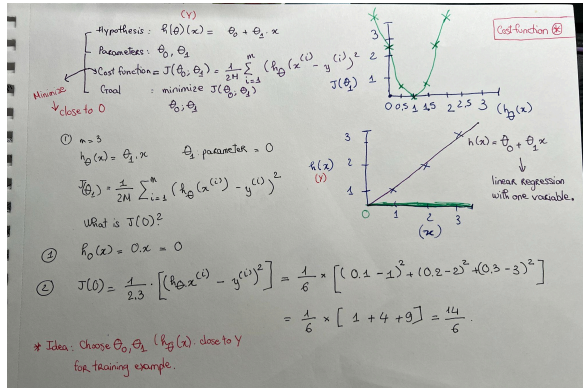
- Before training: weights are random. After training: updated weights enables models to reduce the difference between predicted output and actual target values.

Linear Regression with one variable

- Parametric models: classification and regression.
- We make assumptions about the nature of data distribution:
 - $p(y|x)$ for a supervised problem.
 - $p(x)$ for an unsupervised problem.
- These assumptions: inductive bias (is a statistical model with a fixed number of parameters).

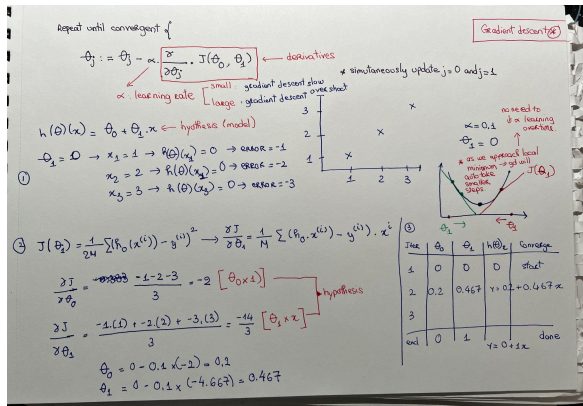
Introduction to ANN Part II

What role does the cost function play in training a model? Using linear regression as an example, explain how minimizing the cost function helps the model learn from data.



Cost function: measure how close the model's predictions are to actual training data. In linear regression, the squared error cost function is minimized to find the parameters that best fit the data.

- Hypothesis: predict the output using two parameters (θ_0 and θ_1)
- θ_0 : removes the intercept parameter.
- Cost function's objective: minimize this function.
- The goal: find parameters (θ_0 and θ_1) that minimize the cost function.



Question 19 (Calculation)

Suppose

$$\theta_1 = 2$$

$$\alpha = 0.1$$

Compute the next value of θ_1 .

Answer

$$\theta_1 = 2 - 0.1(4)$$

$$\theta_1 = 2 - 0.4$$

$$\theta_1 = 1.6$$

Learning Rate: control the step size used when updating parameters during gradient descent. Why must parameters update simultaneously during gradient descent? Simultaneous updates ensure each parameter updates the same previous parameter values, preventing incorrect updates. If learning is too small, convergence becomes very slow. If learning is too fast, the algorithm may overshoot the minimum and fail to converge.

Harder Short Answer Question

1. **Why the cost function for linear regression (bowl-shaped surface) when plotted against θ_0 and θ_1 ?** As the cost function uses squared error terms, it produces a convex quadratic surface. This results in a bowl-shaped cost surface with a single global minimum.
2. **What is a contour plot and why is it used when visualizing cost functions?** A contour plot shows level curves of equal cost values on the cost surface. It allows visualization of the optimization landscape in two dimensions when the cost surface is three-dimensional.
3. **If three parameter points have the same cost value, how are they represented on a contour plot?** They lie on the same contour line, meaning the cost function $J(\theta_0, \theta_1)$ is identical at those points.
4. **In gradient descent, what determines the direction of parameter updates?** The direction is determined by the negative gradient of the cost function, which points toward the direction of steepest decrease.
5. **Suppose the derivative is positive. What will happen to θ_0 during gradient descent?** $\theta_0 = \theta_0 - \alpha \cdot \text{derivative}$. If the derivative is positive, θ_0 decreases.
6. **Explain why gradient descent may end at different local minima depending on the starting point.** When the cost surface has multiple valleys, the algorithm follows the slope from its starting point and may converge to different local minima depending on the initial parameters.
7. **Why do gradient descent steps become smaller near the minimum?** Near the minimum, the gradient approaches zero. Therefore parameter updates become smaller automatically.
8. **If gradient descent oscillates and never converges, what is the most likely cause?** The learning rate α is too large, causing the algorithm to overshoot the minimum repeatedly.